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Land Security and Mobility Frictions

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1 Big picture

- **Question.** How much of China’s slow labor reallocation out of agriculture is due to insecure rural land rights, and how much is due to all other labor-mobility barriers such as hukou restrictions, migration costs, or other labor-market frictions?
- **Core idea.** In rural China, a family that rents out land may fear losing its use rights or compensation if land is later reallocated or expropriated. That fear makes a household keep someone farming the land, even when the family would otherwise prefer to move labor into nonagriculture. Land insecurity therefore behaves like a *household-level migration barrier*.
- **Main conceptual contribution.** The paper builds a Roy-style structural model with *families*, not just individuals. One family member must operate the farm if the household wants to retain land. This creates within-household selection: the family chooses who farms, who migrates, and who works part-time.
- **Data and setting.** Rural China, primarily 2004–2018, using detailed individual- and household-level panel data from the RCRE, aggregate employment data, and a supplementary survey measuring perceived land insecurity.
- **Main 2004 result.** Eliminating land insecurity reduces the share of village families operating farms from 73.6% to 28.2%, lowers the agricultural employment share among villagers from 56.4% to 45.6%, and raises agricultural labor productivity by 24.7%.
- **Comparison with other mobility barriers.** Eliminating residual labor-mobility barriers generates a similar drop in agricultural employment, but a much smaller selection effect: village families operating farms only fall to 49.1%, and agricultural labor productivity rises by 15.9%.
- **Time-series result.** Between 2004 and 2018, the paper finds that *all* of the decline in overall mobility barriers comes from improved land security. Residual labor-mobility barriers actually rise slightly.
- **Amplification result.** Land security interacts strongly with within-agriculture misallocation. If farm-level distortions are also removed, land security produces much larger productivity gains than when introduced in isolation.
- **Why this paper matters.** The paper shifts the development-policy discussion away from seeing migration frictions only through the hukou lens. Rural property rights can matter just as much for structural transformation because they affect both land allocation and family labor allocation.

2 Data and measurement (institutional context and key objects)

2.1 Institutional background: land and labor institutions in China

- **Land institutions.** Under the Household Responsibility System, rural households received *use rights* and residual income rights over land, but not full ownership. Land remained collectively owned by the village.
- **Egalitarian allocation.** Village land was allocated in a highly egalitarian way, tied to village membership through hukou. As a result, within-village landholdings were only weakly related to farming ability.

- **Reallocation risk.** Through the late 1990s, village officials often carried out village-wide land reallocations. Households that rented out land feared that those rentals would be interpreted as evidence that they did not need the land.
- **“Use it or lose it.”** Thin rental markets were partly driven by the fear that renting out land could lead to loss of land rights or loss of compensation if the state later expropriated the land.
- **Land reforms.** The 1998 Land Management Law, the 2003 Rural Land Contracting Law, later transfer reforms, and the 2018 land titling push all strengthened effective land security without privatizing farmland.
- **Labor institutions.** The hukou system restricted movement from rural to urban areas and limited migrants’ access to local public services. Reforms relaxed some restrictions, but large-city access remained difficult and rising urban housing prices likely raised effective migration costs.

2.2 Main data sources

- **RCRE rural household survey.** The core micro data come from the Research Center for the Rural Economy under China’s Ministry of Agriculture.
- **Coverage.** For 2004–2018, the unbalanced panel contains more than 20,000 households per year from roughly 300 villages nationwide.
- **Individual detail.** Since 2003 the survey includes individual labor-supply information, which is crucial because the model needs to know who farms, who works off-farm, and who splits time across sectors.
- **Farm production variables.** The survey contains farm inputs, outputs, land rental activity, household composition, and sector-specific incomes.
- **Aggregate data.** The paper supplements the micro data with aggregate statistics from the Chinese Statistical Yearbook and revised employment series.
- **Supplementary land-insecurity survey.** A separate survey asks farmers about the historical frequency of reallocations and the perceived probability of losing land they rent out. This is what allows the paper to discipline land insecurity directly instead of treating it as a purely latent wedge.

2.3 Stylized facts that motivate the model

- **Rapid structural change after 2004.** Nationwide agricultural employment falls from 39.1% in 2004 to 13.7% in 2018.
- **Two forces behind this fall.** The share of households living in rural areas declines from 69.1% to 39.1%, and among rural residents the share involved in agriculture falls from 69.8% to 44.2%.
- **Composition of rural work.** Full-time nonagricultural work among rural individuals rises from 30.2% to 55.8%, while part-time work falls from 20.7% to 12.0%.
- **Farm size remains tiny.** Average farm size rises from only 0.59 hectares in 2004 to 0.91 hectares in 2018. Even after substantial structural change, farms remain extremely small.

- **Rental-market activity rises mainly on the intensive margin.** The share of farm households renting in land moves only slightly, from 17.9% to 18.7%, but the share of land rented in rises from 14.4% to 31.8%.
- **Before 2004 the transition was slower.** Between 1995 and 2002 agricultural employment falls from 48.0% to 42.2%, and average farm size is essentially unchanged. That slower pre-2004 transition helps motivate the paper's emphasis on post-2003 land reforms.

2.4 Key empirical objects used for calibration

- **Share of village families operating farms.** This is the core family-level moment used to identify the intensity of land insecurity.
- **Sectoral income gap.** The gap between farming profit and nonagricultural wage income is the key individual-level moment used to identify residual labor-mobility barriers.
- **Dispersion in farm productivity and distortions.** Farm TFP and farm TFPR moments are used to discipline ability dispersion and residual agricultural misallocation.
- **Within-family correlations.** Correlations of wages and profits across household members are critical because the model explicitly features family-level sorting.
- **Part-time labor allocation.** The share of part-time workers and the fraction of their time spent in agriculture pin down the model's part-time work technology and fixed cost.

3 Models and empirical specifications (all equations + interpretation)

3.1 Preferences and nonagricultural production

The model uses non-homothetic preferences,

$$u(c_a, c_n) = \phi \log(c_a - \bar{a}) + (1 - \phi) \log(c_n), \quad \phi \in (0, 1), \quad (1)$$

and a linear nonagricultural production technology,

$$Y_n = A_n H_n. \quad (2)$$

- **Interpretation.** The subsistence term $\bar{a}$ generates income effects and therefore structural transformation: as income rises, households spend a smaller fraction of resources on agriculture.
- **Urban side of the model.** Urban households are homogeneous and work in nonagriculture. The rural side is where all the heterogeneity and mobility frictions live.
- **Wage implication.** In the competitive nonagricultural sector, the wage per efficiency unit is simply $w_n = A_n$.

3.2 Agricultural production and land rental

Each village family has J members. If member j operates the family farm, output is

$$y_{ij} = A_a s_{ij} \left(\ell_{ij}^\theta n_{ij}^{1-\theta} \right)^\gamma, \quad (3)$$

where s_{ij} is farm-operator ability, ℓ_{ij} is land input, n_{ij} is labor input, θ is the land share inside the farm technology, and $\gamma < 1$ captures decreasing returns at the farm level.

- **Interpretation.** The more able the chosen operator, the more productive the farm. Because $\gamma < 1$, more able operators should run larger farms, but not infinitely large ones.
- **Land rights.** Every village family starts with the same allocated use rights $\bar{\ell}$. They can rent land in or out, so operational scale is $\ell_{ij} = \bar{\ell} + \ell_{ij}^{rent}$.
- **Core friction.** If families fear losing rented-out land, then the land rental market will not move land to the best operators as efficiently as it should.

3.3 Land insecurity and the farm-operator problem

The farm operator solves

$$\pi(s_{ij}, \tau_{ij}) = \max_{n_{ij}^d, \ell_{ij}^{rent}} \left\{ \tau_{ij} p_a A_a s_{ij} \left[\left(\bar{\ell} + \ell_{ij}^{rent} \right)^\theta (n_{ij}^d)^{1-\theta} \right]^\gamma - q \ell_{ij}^{rent} - w_a n_{ij}^d + \eta \phi_i \ell_{ij}^{rent} \mathbf{1}[\ell_{ij}^{rent} < 0] \right\}. \quad (4)$$

- **Meaning of η .** η is the probability that a household loses the use rights to land it rents out.
- **Meaning of ϕ_i .** ϕ_i is the family-specific perceived income loss per unit of rented-out land if that loss occurs.
- **Why the last term is a cost.** For a household that rents out land, $\ell_{ij}^{rent} < 0$, so the final term is negative. It is the expected penalty from insecure land rights.
- **Role of τ_{ij} .** τ_{ij} is a residual farm-level distortion that captures remaining within-agriculture wedges beyond the land insecurity channel.

3.4 Individual occupation incomes and part-time work

An individual working full-time in nonagriculture earns

$$i_{ij}^{FN} = w_n h_{ij} (1 - \xi_{ij}), \quad (5)$$

while a full-time agricultural worker earns

$$i_{ij}^{FA} = w_a. \quad (6)$$

Part-time work is allowed in both sectors. If n_{ij} is labor supplied to agriculture and c is a fixed cost of working part-time, then optimal agricultural labor supply for a part-time worker is

$$n_{ij}^* = \min \left\{ \left(\frac{\nu \kappa w_a}{h_{ij} w_n (1 - \xi_{ij})} \right)^{\frac{1}{1-\nu}}, 1 - c \right\}, \quad (7)$$

with part-time income

$$i_{ij}^{PT} = h_{ij} w_n (1 - c - n_{ij}^*) (1 - \xi_{ij}) + w_a \kappa (n_{ij}^*)^\nu. \quad (8)$$

Effective labor supply to agriculture is then

$$n_{ij} = \begin{cases} 1, & \text{if } i_{ij}^{FA} \geq i_{ij}^{FN} \text{ and } i_{ij}^{FA} \geq i_{ij}^{PT}, \\ 0, & \text{if } i_{ij}^{FN} \geq i_{ij}^{FA} \text{ and } i_{ij}^{FN} \geq i_{ij}^{PT}, \\ \kappa (n_{ij}^*)^\nu, & \text{if } i_{ij}^{PT} \geq i_{ij}^{FA} \text{ and } i_{ij}^{PT} \geq i_{ij}^{FN}. \end{cases} \quad (9)$$

- **Interpretation of ξ_{ij} .** ξ_{ij} is an individual-specific wage barrier to nonagriculture. It is the paper's catchall for all mobility costs other than land insecurity.
- **Why part-time work matters.** In the Chinese micro data, part-time sectoral work is common. The model therefore needs an interior margin, not just a binary agriculture-versus-nonagriculture choice.
- **Economic logic.** Land insecurity works through the household's farming decision; ξ_{ij} works through each individual's outside option.

3.5 Household allocation problem

If member j is assigned to be the operator, household income is

$$I_i^a(\text{operator} = j) = \pi(s_{ij}, \tau_{ij}) + \sum_{k \neq j} \max\{i_{ik}^{FN}, i_{ik}^{FA}, i_{ik}^{PT}\}. \quad (10)$$

The household chooses the best operator,

$$I_i^a = \max_{j \in J} I_i^a(\text{operator} = j). \quad (11)$$

If the family does *not* farm, it rents out all allocated land and earns

$$I_i^n = \sum_j \max\{i_{ij}^{FN}, i_{ij}^{FA}, i_{ij}^{PT}\} + (1 - \eta) q \bar{\ell} - \eta \phi_i \bar{\ell}. \quad (12)$$

Total household income is

$$I_i = \max\{I_i^a, I_i^n\}. \quad (13)$$

- **Most important margin in the paper.** The choice between I_i^a and I_i^n is what makes land insecurity a *family-level* barrier.
- **Intensive margin.** Land insecurity discourages farm operators from renting out additional land.
- **Extensive margin.** Land insecurity discourages the family from shutting down the farm entirely, because doing so requires renting out all allocated land and taking the risk of loss.
- **Selection implication.** Families may keep low-opportunity-cost members in farming, even when those individuals are not the most productive farm operators. That is the within-household selection mechanism.

3.6 Ability distributions and parametric wedges

The paper allows ability to have common family components and individual-specific components:

$$\log(h_{ij}) = \log(h_i^H) + \log(h_{ij}^I), \quad (14)$$

$$\log(s_{ij}) = \log(s_i^H) + \log(s_{ij}^I) + \lambda \log(h_{ij}). \quad (15)$$

Labor-mobility barriers are parameterized as

$$\xi_{ij} = \frac{\exp(\mu_\xi + \varepsilon_{ij}^\xi)}{1 + \exp(\mu_\xi + \varepsilon_{ij}^\xi)}, \quad (16)$$

and residual farm distortions as

$$\log(\tau_{ij}) = \zeta \log(s_{ij}) + \varepsilon_{ij}^\tau. \quad (17)$$

Finally, family-specific land insecurity is distributed as

$$\log(\phi_i) \sim \mathcal{N}(\mu_\phi, \sigma_\phi^2). \quad (18)$$

- **Interpretation.** The model allows comparative advantage, family-level correlation in ability, and systematic correlation between farm ability and farm distortions.
- **Why this matters.** The paper's quantitative results depend heavily on who is likely to farm and who is likely to migrate. Those decisions cannot be matched with a one-dimensional ability draw.

4 Identification and instruments (what makes the estimates credible)

4.1 No external IV: identification comes from distinct household and individual margins

- **Key identification insight.** Land insecurity is a *family-level* friction because it changes whether the family keeps a farm operator. Residual mobility barriers are *individual-level* frictions because they govern whether particular household members work in nonagriculture.
- **Empirical implication.** The share of village families operating farms is highly sensitive to land insecurity, while the sectoral income gap is highly sensitive to labor-mobility barriers.
- **Why the decomposition works.** Only one family member is needed to keep the household farming. So even if land insecurity keeps the family attached to agriculture, the remaining members still respond to individual mobility costs. This separation is what lets the model distinguish the two wedges.

4.2 How the paper identifies land insecurity

- **Direct measurement of η .** The supplementary survey measures two objects: the historical frequency of reallocations/land takings and the perceived conditional chance of losing rented-out land in those events.
- **Numerical discipline.** Up to 2004, the probability that land was affected by reallocation or land taking was 10.2%. The perceived conditional loss probability was 50%. Their product gives $\eta = 5.1\%$.
- **Mean land-insecurity cost.** The average loss parameter μ_ϕ is pinned down by the observed share of village families operating farms in 2004, about 74%.
- **Dispersion in land insecurity.** σ_ϕ is identified from selection: it helps the model match wage differences between households that do and do not operate farms.
- **Economic plausibility check.** The calibrated income loss from losing land is around five times annual rural household income, which the authors argue is consistent with the discounted present value of future agricultural income flows.

4.3 How the paper identifies labor-mobility barriers and abilities

- **Mean labor barrier.** μ_ξ is chosen to match the sectoral gap between farming profit and nonagricultural wage income.
- **Dispersion in labor barriers.** σ_ξ is chosen to match the rank correlation between nonagricultural wages and labor supply among part-time workers.
- **Family ability in nonagriculture.** Wage dispersion and within-family wage correlations identify the family and idiosyncratic components of nonagricultural ability.
- **Farm ability.** The dispersion of farm TFP identifies the spread of farm-operator ability, while the correlation between nonagricultural wage income and farm profit helps identify cross-sector ability correlation λ .

- **Farm distortions.** The dispersion of TFPR and the correlation of TFP with TFPR among households active in rental markets identify the farm-wedge parameters.

4.4 What makes the decomposition convincing

- **Model fit.** The paper matches 14 empirical moments closely, including employment shares, part-time labor allocation, within-family wage dependence, farm productivity dispersion, and the nominal agriculture-versus-nonagriculture productivity ratio.
- **Sharp falsification exercise.** If the model sets land insecurity to zero and tries to explain the sectoral income gap only with labor-mobility barriers, it predicts that only 42.4% of village families operate farms, far below the observed 73.7%.
- **Black-box point.** Without land insecurity, the model would need average labor-mobility barriers of 79.5% to match the sectoral gap, versus 54.0% in the baseline. So a standard migration-cost interpretation would wildly overstate the role of non-land frictions.
- **Out-of-sample rental-market discipline.** In the data, 78% of farm households do not participate in the rental market; the calibrated model implies 73%. Among households renting in land, the model’s average rental amount is close to the data as well.

5 Tables and figures

5.1 Table I: Employment in agriculture

Read:

- The table shows that nationwide agricultural employment collapses from 39.1% in 2004 to 13.7% in 2018.
- About half of that decline comes from the shrinking rural population and half from reallocation *within* the rural population from agriculture to nonagriculture.
- Among rural individuals, the largest change is the rise in full-time nonagricultural work from 30.2% to 55.8%. That is exactly why the paper needs a model of within-household labor allocation rather than a model of farm households only.

TABLE I
EMPLOYMENT IN AGRICULTURE

Variable	2004	2009	2013	2018
Nationwide agricultural employment share (%)	39.1	28.4	20.9	13.7
Share of all households living in rural area (%)	69.1	57.2	46.4	39.1
Share of labor days in agriculture among rural households (%)	56.5	49.6	45.1	35.1
Share of individuals in rural areas involved in agriculture (%)	69.8	62.0	56.7	44.2
Share of farm operators (%)	27.6	23.8	21.9	16.7
Share of full-time workers (%)	21.5	20.3	18.7	15.5
Share of part-time workers (%)	20.7	17.9	16.1	12.0
Share of individuals in rural areas in full-time nonagriculture (%)	30.2	38.0	43.3	55.8

Notes. Nationwide statistics adjusted from the Chinese Statistics Yearbook. All other statistics are calculated from micro-level data from the RCRE survey, as shares out of the total numbers of employed individuals in the survey. For each year the shares of farm operators, full-time, and part-time workers sum up to the total share in agriculture, save for rounding.

TABLE II
AVERAGE FARM SIZE AND LAND RENTALS

Variable	2004	2009	2013	2018
Average farm size (ha)	0.59	0.62	0.71	0.91
Share of households renting in land (%)	17.9	20.0	17.9	18.7
Share of land rented in (%)	14.4	22.4	18.6	31.8

Notes. Average farm size is the total cultivated land over the total number of operating farms (households) from the RCRE survey data and expressed in hectares (ha). Statistics on land rentals are calculated among farming households in the survey each year. Data are from authors’ calculation from the micro-data sample.

5.2 Table II: Average farm size and land rentals

Read:

- Farm size rises from 0.59 to 0.91 hectares, so there is some reallocation, but farms remain extremely small.
- The extensive margin of renting barely moves, while the intensive margin rises a lot. This is consistent with gradual easing of land-market frictions rather than a fully liquid rental market.
- The table supports the paper’s interpretation that land reform matters, but also that land-market distortions remain substantial.

5.3 Table III: Structural transformation before 2004

Read:

- Before 2004 structural change is much slower: agricultural employment falls only from 48.0% to 42.2% between 1995 and 2002.
- Farm size barely changes before 2004. The contrast with the post-2004 period fits the story that land reforms in the early 2000s materially improved the reallocation environment.

Variable	1995	2002
Nationwide agricultural employment share (%)	48.0	42.2
Share of households living in rural area (%)	75.4	72.6
Share of labor days in agriculture among rural households (%)	63.6	58.2
Share of individuals in rural areas involved in agriculture (%)	73.6	67.5
Share of full-time workers (%)	53.6	48.8
Share of part-time workers (%)	19.9	18.8
Average farm size (ha)	0.57	0.59

Notes. Nationwide statistics adjusted from the Chinese Statistics Yearbook. Employment statistics are calculated from micro-level data from the RCRE survey for 1995 and 2002, as shares out of the total numbers of employed individuals in the survey. Average farm size is the total amount of farm land over the total number of operating farms (households). The numbers are not directly comparable to those in the survey over 2004–2018.

Moments	Data	Model
Employment share among village individuals:		
Full-time nonagriculture	0.302	0.299
Part-time	0.207	0.204
Median fraction of part-time hours in agriculture	0.286	0.286
Rank correlation of wages and part-time hours in nonagr.	0.398	0.397
Share of village households with farm operators	0.737	0.736
Sectoral gap: nonagr. wage versus farming profit	0.081	0.081
Family wage differentials:		
Average nonagr. wage, with/without operators	−0.281	−0.282
Wage dispersion among full-time nonagr. workers:		
Standard deviation	0.610	0.608
Within-family correlation	0.558	0.557
Correlation of nonagricultural wage income and farm profit	0.080	0.078
Agricultural production:		
Standard deviation of farm TFP	0.657	0.660
Standard deviation of farm TFPR	0.633	0.631
Rank correlation of farm TFP and TFPR	0.968	0.975
Nominal agr. to nonagr. labor productivity ratio	0.388	0.389

Note. Reporting the 14 moments targeted in the data and the corresponding model estimated moments.

5.4 Table IV: Targeted moments, data versus model

Read:

- This is the calibration discipline table. The model matches full-time nonagricultural employment, part-time work, the share of households with operators, wage dispersion, farm TFP dispersion, farm TFPR dispersion, and the agriculture/nonagriculture labor productivity ratio.
- The most important moments for the paper’s identification are the share of village households operating farms (0.737), the sectoral income gap (0.081), and the rank correlation between farm TFP and TFPR (0.968).
- Because the fit is tight on both household-level and individual-level moments, the counterfactual decomposition of land insecurity versus labor-mobility barriers is not coming from one loose

moment but from a whole joint set of restrictions.

5.5 Table V: The role of land security and labor-mobility barriers

Read:

- This is the central quantitative table of the paper.
- Eliminating land insecurity turns many farms into exits: the share of village families operating farms falls from 73.6% to 28.2%. The authors interpret many baseline farms as “zombie” farms maintained mainly to protect land rights.
- Agricultural labor productivity rises by 24.7% under land security, compared with 15.9% when other labor-mobility barriers are removed.
- Removing labor-mobility barriers has a similar effect on agricultural employment but much weaker effects on operator selection. That is the key distinction between household-level and individual-level frictions.
- The nominal agricultural productivity gap falls from 2.57 to 2.20 with land security and to 1.39 with no mobility barriers. So the two frictions matter differently for real productivity and nominal sectoral gaps.

TABLE V
THE ROLE OF LAND SECURITY AND LABOR-MOBILITY BARRIERS

	Baseline	Land security	No labor barriers
Village families operating farms (%)	73.6	28.2	49.1
Agricultural employment share among villagers (%)	56.4	45.6	46.3
Δ Agricultural output (%)	—	+0.9	−4.9
Δ Agricultural labor productivity (%)	—	+24.7	+15.9
Δ Median log farm-operator ability (%)	—	+22.1	+3.9
Δ Nonagricultural output (%)	—	+4.0	+4.6
Δ Real GDP per capita (%)	—	+3.5	+2.7
Within-household selection in farming:			
% of farm operators with highest s_{ij}	61.2	69.6	62.6
Nominal agricultural productivity gap	2.57	2.20	1.39

Notes: Statistics for the baseline model, counterfactual land security ($\eta_{ij} = 0$ for all i), and counterfactual no labor-mobility barriers ($\xi_{ij} = 0$). The first two and the last two statistics are in levels, while all other statistics are displayed as differences compared to the baseline model.

TABLE VI
THE EVOLUTION OF FRICTIONS OVER TIME

	Baseline	2018 land security	2018 labor barriers
Village families operating farms (%)	73.6	37.6	83.5
Agricultural employment share among villagers (%)	56.4	47.3	61.8
Δ Agricultural output (%)	—	+1.1	+3.3
Δ Agricultural labor productivity (%)	—	+20.6	−5.7
Δ Median log farm-operator ability (%)	—	+14.6	−0.4
Δ Nonagricultural output (%)	—	+3.4	−2.3
Δ Real GDP per capita (%)	—	+3.1	−1.5
Within-household selection in farming:			
% of farm operators with highest s_{ij}	61.2	66.6	63.4
Nominal agricultural productivity gap	2.57	2.32	3.41

Notes: Statistics for the baseline 2004 economy and two counterfactuals with frictions at their 2018 value. We replace the level of land frictions η and μ_{φ} such that the average $\left(\frac{\eta_{ij}}{I_i}\right)$ matches that of the 2018 level (second column) and labor-mobility frictions ξ_{ij} to 2018 values (third column). The first two and the last two statistics are in levels, while all other statistics are displayed as differences compared to the baseline 2004 economy.

5.6 Table VI: The evolution of frictions over time

Read:

- Replacing 2004 land insecurity with its estimated 2018 level lowers the farm-operator household share from 73.6% to 37.6% and raises agricultural labor productivity by 20.6%.
- Replacing 2004 labor-mobility barriers with their estimated 2018 level actually *raises* agricultural employment and lowers agricultural labor productivity.
- The punchline is stark: observed structural change over 2004–2018 is accounted for by improved land security, not by falling residual migration barriers.

5.7 Table VII: Interaction between land security and misallocation within agriculture

Read:

- This table adds another key insight: land security and within-agriculture reform are complements.
- Eliminating idiosyncratic farm distortions alone raises agricultural labor productivity by 140%. Combining that reform with land security raises it by 424.5%.
- The mechanism is both extensive and intensive. With secure land and no idiosyncratic farm distortions, low-productivity farmers exit and land flows more effectively toward the highest-ability operators.
- This table directly links the present paper to Adamopoulos et al. (2022): removing static farm misallocation becomes much more powerful once the household-selection margin is also freed.

TABLE VII
INTERACTION BETWEEN LAND SECURITY AND MISALLOCATION WITHIN AGRICULTURE

	Baseline	Land security	No idio. farm dist.	+ Land security
Village families operating farms (%)	73.6	28.2	56.5	2.2
Agr. employment share among villagers (%)	56.4	45.6	36.9	20.2
Δ Agricultural output (%)	—	+0.9	+56.9	+87.7
Δ Agricultural labor productivity (%)	—	+24.7	+140.0	+424.5
Δ Median log farm-operator ability (%)	—	+22.1	−1.9	+161.3
Δ Nonagricultural output (%)	—	+4.0	+6.0	+10.6
Δ Real GDP per capita (%)	—	+3.5	+10.6	+17.2
Within-household selection in farming:				
% of farm operators with highest s_{ij}	61.2	69.6	52.3	93.2
Nominal agricultural productivity gap	2.57	2.20	5.70	3.75

Notes. Statistics of the baseline model for three counterfactual experiments: land security ($\eta_{ij} = 0$), no idiosyncratic farm distortions ($\tau_{ij} = 1$), and both. Statistics for deviations Δ are from the baseline model.

TABLE VIII
STATUS AND LOCATION OF NONAGRICULTURAL WORK BY RURAL INDIVIDUALS

Variable	2004	2009	2013	2018
Share full-time nonagriculture (%)	30.2	38.0	43.3	55.8
Rural (%)	14.8	17.1	20.6	29.6
Urban (%)	15.4	21.0	22.7	26.2
Share part-time nonagriculture (%)	20.7	17.9	16.1	12.0
Rural (%)	15.9	13.7	12.3	9.0
Urban (%)	4.8	4.2	3.7	3.0

Notes. All statistics are calculated from micro-level data from the RCRE survey, as shares out of the total numbers of employed individuals in the survey. For each year the shares of rural and urban workers sum up to the total share in nonagriculture by full- and part-time status.

5.8 Table VIII: Status and location of nonagricultural work

Read:

- More than half of rural nonagricultural work is local rather than urban. In 2004, rural full-time nonagriculture is split roughly evenly between rural and urban locations; by 2018 rural local work remains very important.
- This is why the extension with rural and urban nonagricultural sectors matters: reallocation out of farming is not the same thing as migration to big cities.

5.9 Table IX: Extensions with rural nonagriculture and age differences

Read:

- The main qualitative results survive the richer models. Land security continues to have the largest effect on the fraction of households operating farms and on agricultural selection.
- In the rural-versus-urban nonagriculture extension, eliminating labor-mobility barriers produces especially large gains in nonagricultural output and real GDP per capita, because it now frees movement across both sector and space.
- In the age extension, land security and labor barriers both matter a lot, but mobility-barrier removal is especially important for aggregate output because older individuals face large barriers to leaving agriculture.

TABLE IX
EXTENSIONS

	Baseline	Land security barriers	No labor barriers
Panel A: Rural and Urban Nonagriculture			
Village families operating farms (%)	73.6	27.7	47.4
Agricultural employment share among villagers (%)	55.3	46.6	46.1
Δ Agricultural output (%)	—	−0.2	−5.6
Δ Agricultural labor productivity (%)	—	+18.4	+13.1
Δ Median log farm-operator ability (%)	—	+16.6	+0.4
Δ Nonagricultural output (%)	—	+3.5	+9.8
Δ Real GDP per capita (%)	—	+2.9	+6.6
Panel B: Age Differences			
Village families operating farms (%)	73.6	31.9	44.7
Agricultural employment share among villagers (%)	56.7	46.1	43.4
Δ Agricultural output (%)	—	+1.1	−5.6
Δ Agricultural labor productivity (%)	—	+24.4	+23.3
Δ Median log farm-operator ability (%)	—	+19.6	+5.9
Δ Nonagricultural output (%)	—	+4.0	+10.3
Δ Real GDP per capita (%)	—	+3.6	+6.8

Note. Statistics for the baseline estimated model versus our two counterfactual experiments, land security ($\eta_{pi} = 0$ for all i) and no labor-mobility barriers ($\xi_{ij}^L = \xi_{ij}^H = 0$). Panel A shows results for the extended model with rural and urban nonagriculture, and Panel B for the extended model with rural and urban nonagriculture and young-old age differences. The percentage of village households operating farms and the sectoral employment shares among villagers are in levels, while all other statistics are displayed as differences compared to the baseline model.

TABLE X
OCCUPATION DIFFERENCES BY AGE GROUPS IN THE MODEL

	Baseline	Land security	No labor barriers	Both
Old				
Agricultural employment share (%)	66.3	51.7	37.7	30.8
Operator	31.3	13.2	15.4	8.7
Worker	35.0	38.5	22.3	22.1
Young				
Agricultural employment share (%)	38.6	27.5	30.0	24.4
Operator	19.9	8.9	14.6	9.1
Worker	18.6	18.6	15.3	15.3

Notes. The share of old and young adults among village households in agricultural employment (and between operator and agricultural worker) for the baseline model and counterfactuals.

TABLE XI
MODEL WITH REGIONAL HETEROGENEITY

	Baseline	Land security barriers	No labor barriers
Panel A: Peri-urban			
Village households operating farms (%)	57.2	27.8	33.2
Agricultural employment share among villagers (%)	48.2	40.9	37.0
Δ Agricultural output (%)	—	+1.3	−6.7
Δ Agricultural labor productivity (%)	—	+19.2	+21.4
Δ Median log farm-operator ability (%)	—	+21.1	+9.4
Δ Nonagricultural output (%)	—	+2.6	+6.8
Δ Real GDP per capita (%)	—	+2.4	+4.3
Panel B: Remote			
Village households operating farms (%)	77.8	36.4	52.2
Agricultural employment share among villagers (%)	60.0	51.5	48.8
Δ Agricultural output (%)	—	+0.4	−5.6
Δ Agricultural labor productivity (%)	—	+17.1	+16.2
Δ Median log farm-operator ability (%)	—	+14.2	+2.5
Δ Nonagricultural output (%)	—	+3.6	+9.1
Δ Real GDP per capita (%)	—	+3.1	+5.6

Note. Statistics for the baseline estimated model versus our two counterfactual experiments, land security ($\eta_{pi} = 0$ for all i) and no labor-mobility barriers ($\xi_{ij}^L = \xi_{ij}^H = 0$). Panel A shows results for the peri-urban area and Panel B shows results for the remote area. The percentage of village households operating farms and the sectoral employment shares among villagers are in levels, while all other statistics are displayed as differences compared to the baseline model.

5.10 Table X: Occupation differences by age groups

Read:

- In the baseline, 66.3% of older rural individuals work exclusively in agriculture, compared with 38.6% of younger individuals.
- Removing both land insecurity and labor-mobility barriers leaves an age gap of only 6.3 percentage points, down from 27.8 points.
- The decomposition is useful: about 20.1 percentage points of the age gap come from labor-mobility barriers, and about 3.7 from land insecurity. Land insecurity matters more for who becomes the operator than for overall agricultural attachment.

5.11 Table XI: Model with regional heterogeneity

Read:

- Peri-urban villages start with fewer farming households and lower agricultural employment than remote villages.
- Land insecurity is much less severe near urban areas: securing land lowers the operator-household share by 19.4 percentage points in peri-urban areas, but by 41.4 points in remote villages.
- Local rural nonagricultural mobility barriers are also lower in peri-urban places. The paper therefore highlights meaningful regional heterogeneity inside rural China, not just a national average friction.

6 Short summary

- **Conclusion.** Land insecurity in rural China is not just a land-market distortion; it is also a major labor-mobility barrier because it keeps families attached to farming.
- **Intuition.** A household may keep someone on the farm not because farming is efficient, but because abandoning the farm risks losing the land. That makes the wrong families and the wrong family members stay in agriculture.
- **Empirical lesson.** Distinguishing family-level from individual-level frictions is essential. Looking only at sectoral wage gaps would incorrectly assign too much of the problem to generic migration barriers.
- **Quantitative lesson.** In the paper’s accounting, improved land security explains the reduction in labor-reallocation barriers in China after 2004; residual non-land mobility frictions did not fall.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- **Main finding.** Insecure rural land rights are quantitatively as important as all other labor-mobility barriers combined for agricultural employment and agricultural productivity in China circa 2004.
- **Magnitude.** Secure land rights reduce the fraction of village households operating farms by more than 45 percentage points and raise agricultural labor productivity by roughly one quarter.
- **Dynamic conclusion.** The improvement in labor reallocation between 2004 and 2018 is fully accounted for by better land security, not by falling residual migration barriers.
- **Selection contribution.** The paper’s biggest conceptual value is to show that land institutions distort not only factor allocation across farms but also talent allocation across households, age groups, and locations.

- **Policy implication.** Rural property-rights reform can be a first-order development policy. Reducing hukou-style barriers alone is not enough if households still fear losing land when they leave farming.

7.2 Economic intuition

- If land were secure, low-productivity rural families would rent out land and move labor into nonagriculture, while high-ability operators would expand.
- Under insecure land rights, land does not fully move to the best farmers and many low-productivity families keep a minimal farming presence as insurance.
- Because the family chooses *who* farms, land insecurity also distorts within-household assignment. The operator may simply be the person with the weakest outside option rather than the strongest farming skill.
- That is why secure land rights improve both static allocation and selection, and why the gains are especially large when combined with reforms that remove residual farm-level distortions.

7.3 Takeaway

This paper is a strong example of how a property-rights institution can enter a structural transformation problem through a non-obvious channel. In the paper’s framework, land insecurity is not merely about misallocated acreage; it is about households using farming to defend claims on an asset. Once that mechanism is built into a family-level model, the data imply that better land security was central to China’s post-2004 reallocation out of agriculture.